Multiyear Sentinel-1/2 and CHIRPS Integration for Flood Prediction and Agricultural Exposure Assessment in the Senegal River Valley

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Abstract

Floods threaten agriculture in the Senegal River Valley, but flood mapping, cropland exposure, and prediction require distinct evidence. We analyze 2019–2024 Google Earth Engine exports combining Sentinel-1, Sentinel-2, CHIRPS, historical water indicators, and 200 repeated sites. The point–period table has 19,200 records; 13,583 have a valid next-period target, of which 3.76% are positive. We compare five classifiers using site-grouped spatial validation, a 2024 temporal hold-out, and a joint future-and-unseen-site test. After excluding a dry-reference feature that overlaps with label construction, Random Forest is selected by spatial validation (mean average precision 0.220, SD 0.214); its 2024 temporal and joint space–time average precision are 0.669 and 0.379, respectively. SHAP highlights radar backscatter, location, and historical water extent. These labels remain Sentinel-1-derived proxies. The supplied agricultural-exposure rasters are rendered RGB images without numeric class masks, and no field damage or yield data are available; crop exposure and production loss therefore cannot be quantified from these exports.

Keywords: Flood prediction, agricultural exposure, Sentinel-1, Sentinel-2, CHIRPS, Dynamic World, Google Earth Engine, Senegal River Valley

1 Introduction

Floods can disrupt agricultural production through direct crop submergence, delayed field operations, erosion, and prolonged soil saturation. In semi-arid agricultural regions, however, the relationship between inundation and crop damage depends on flood duration, crop type, phenological stage, and local water management. Consequently, mapping flooded pixels alone does not quantify agricultural exposure, and agricultural exposure does not by itself prove crop damage or yield loss.

Remote-sensing studies in the Senegal River Valley have examined flood monitoring and flood-recession agriculture for decades [1, 2]. Recent work maps seasonal flood-recession cropland extent in the valley, while field-oriented analyses describe water constraints on flood-recession farming [3, 4]. These studies motivate joint flood and agricultural-exposure assessment but do not make a satellite overlap map equivalent to measured crop damage or yield loss.

Despite this progress, three gaps remain relevant to the Senegal River Valley: (i) the need to examine flood dynamics across multiple seasons rather than a single season; (ii) the need to quantify the spatial overlap between potential inundation and candidate cropland; and (iii) the need for temporally honest prediction evaluation that separates earlier seasons from a later test season. This study addresses these gaps through a reproducible Google Earth Engine (GEE) workflow.

The objectives are to:

1. construct a multiyear, multimodal point–time dataset from Sentinel-1, Sentinel-2, CHIRPS, and historical water information;
2. map potential newly inundated surfaces and summarize their area and temporal frequency;
3. assess which supplied products support quantification of potential cropland exposure and identify missing evidence;
4. compare next-period classifiers using spatial, temporal, and joint space–time validation, with feature attribution.

The term *agricultural exposure* is used deliberately. Without independent agronomic or yield observations, the workflow cannot establish that an exposed field suffered a particular level of damage or production loss.

2 Related work

2.1 Flood mapping from radar and optical imagery

Sentinel-1 flood mapping ranges from threshold and change-detection processing to supervised classifiers and deep segmentation. Automated Sentinel-1 chains have reported event-dependent performance, and reviews emphasize the roles of training labels, sensors, and test-event design [5–7]. U-Net is an established segmentation architecture when spatially labeled training data are available [8].

2.2 Agricultural exposure and crop damage

The relationship between inundation and crop impact depends on crop type, phenology, water management, and flood duration. For this reason, exposure maps should be distinguished from field-confirmed damage and yield outcomes. A relevant regional contribution is the mapping of flood-recession cropland extent in the Senegal River Valley [3]; it is a cropland-mapping study, not a directly comparable next-period flood-classification benchmark.

The most relevant operational forecasting comparison is Google's river-flood system, which uses gauge-stage and rainfall sequences with year-held-out and extreme-event evaluation; its reported inundation-extent F1 values are not the same target as our next-dekad point labels [9]. At global scale, an AI system predicts extreme river floods in ungauged watersheds at up to five days lead time and compares reliability against a global hydrological nowcast, again using a different target and reference framework [10]. These are state-of-the-art context, not directly interchangeable score baselines. Spatially or temporally structured observations require blocked validation rather than random row splits [11].

3 Methods

3.1 Study area, period, and sampling framework

The study focuses on the Senegal River Valley (SRV), Senegal. The point source specified by the project is the pre-existing Earth Engine feature collection `projects/project37246/assets/points_200_vallee`, containing 200 point identifiers and coordinates. The supplied workspace does not contain its generation code or sampling-design documentation; therefore, the selection cannot be characterized as random, systematic, or stratified. Coordinates span -16.4872 to $-12.0523°$ longitude and 14.1457 to 16.8234° latitude. All 200 identifiers recur in each year and each point has a single stable coordinate pair. The analysis covers 1 June–31 October in 2019–2024, represented by 16 ten-day periods per year. Each of the 96 year–period groups contains 200 records, yielding $200 \times 16 \times 6 = 19{,}200$ point–period rows.

The primary key is (`point_id`, `date_debut`); the export contains no duplicate keys. Temporal joins use the point identifier and period start rather than coordinate equality. Coordinate stability was checked within each identifier. The next-period label was compared with the within-point, within-year one-period shift of the current flood label; all 13,583 comparable pairs agree. The original extraction code, source rasters, and resampling parameters are not included, so pixel-level reprojection and the point-generation procedure cannot be independently reconstructed from the CSVs alone.

3.2 Data products and harmonization

The point–period export integrates Sentinel-1 GRD VV/VH backscatter and flood-screen indicators; Sentinel-2 Level-2A-derived NDVI, EVI, MSAVI, and NDWI; daily CHIRPS precipitation summaries; JRC Global Surface Water historical occurrence/extent; and Dynamic World crop-class probabilities used by the original workflow for candidate cropland. The sensors differ in pixel size, acquisition frequency,

Table 1 Input products represented in the supplied export and their analytical roles.

Product	Exported fields	Role and principal limitation
Sentinel-1 GRD	VV, VH, water, dry reference, current/next label	Potential inundation proxy; not an independent flood reference.
Sentinel-2 L2A	NDVI, EVI, MSAVI, NDWI	Vegetation/water context; indices are missing when optical composites are unavailable.
CHIRPS daily	Ten-day, maximum-day, rainy-day and antecedent rainfall; SPI-labelled fields	Coarse precipitation covariates; no station-based local validation is supplied.
JRC Global Surface Water	Historical occurrence and maximum extent	Long-term water context, not an event-specific flood reference.
Dynamic World	Seasonal crop probability in the source workflow	Candidate cropland context; the numeric exposure table/mask is absent from the delivered exports.

and cloud sensitivity: Sentinel-1/2 and Dynamic World are nominally 10 m products, JRC water is approximately 30 m, and CHIRPS is a substantially coarser precipitation grid. The supplied Earth Engine export has already sampled these products to common point–period records, but the exact reprojection and resampling settings are not available. All statistical results below use this delivered table, not a reconstructed raster stack. Product descriptions are available in the source literature [12–17].

3.3 Exploratory statistics and correlation tests

Exploratory analysis includes variable types, unique values, missingness, descriptive statistics, annual and ten-day target prevalence, per-site recurrence conditional on valid observations, and period-level inundated-area summaries. Pearson product–moment (r) and Spearman rank (ρ) correlations are computed pairwise for numeric covariates and each flood target using available pairs. Two-sided nominal p-values are reported, with Benjamini–Hochberg false-discovery-rate q-values across predictor associations for each target. These p-values are descriptive: repeated measurements at the same locations and temporal dependence violate independent-row assumptions, and adjustment for multiple tests does not remove that dependence. Correlation is not interpreted causally. Highly collinear rainfall summaries are retained as candidate predictors but identified in the full correlation output.

3.4 Sentinel-1 observations and potential inundation

Sentinel-1 GRD imagery is obtained from `COPERNICUS/S1_GRD`. Images in IW mode with VV and VH polarizations are filtered to the study region and each ten-day period. Median composites are generated, and a spatial median filter is applied to reduce speckle. The VV backscatter is used for an initial potential-water screen:

$$W_{i,t} = \begin{cases} 1, & VV_{i,t} < \tau_{\mathrm{VV}}, \\ 0, & VV_{i,t} \geq \tau_{\mathrm{VV}}, \end{cases} \tag{1}$$

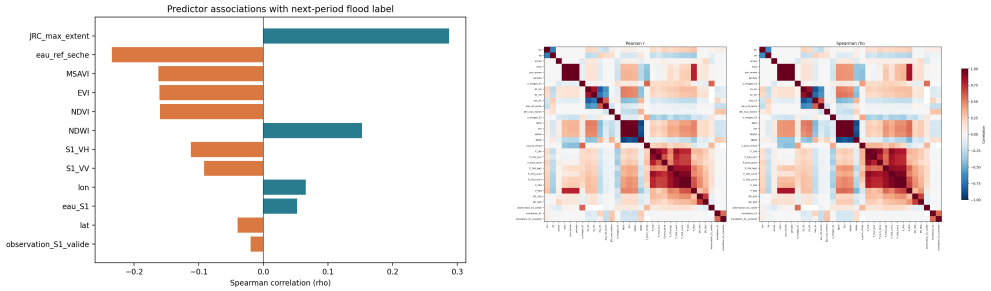


Fig. 1 Spearman associations of predictors with the next-period target (left) and full Pearson/Spearman correlation matrices (right). Pairwise sample sizes, p-values, and false-discovery-rate q-values are supplied in the CSV outputs.

where the initial threshold is $\tau_{VV} = -16$ dB. This is an operational screening threshold, not a universally valid water/land boundary, and it should be tested against independent reference data.

For each year, a pre-season reference is generated from 1 March to 15 May. Let $W_{i,y}^{\text{ref}}$ be the potential-water indicator derived from that reference. Potential newly inundated pixels are defined as

$$F_{i,t} = W_{i,t}\left(1 - W_{i,y}^{\text{ref}}\right). \tag{2}$$

The label $F_{i,t} = 1$ therefore indicates water detected in the current composite but not in the pre-season reference. It is not equivalent to independently confirmed flooding. Pixels without valid radar observations are treated as missing rather than as non-flooded.

The JRC Global Surface Water product (`JRC/GSW1_4/GlobalSurfaceWater`) provides historical water occurrence and maximum extent. These layers are contextual covariates and are not interpreted as event-specific flood observations.

3.5 Sentinel-2 vegetation and water indices

Sentinel-2 Level-2A imagery is obtained from `COPERNICUS/S2_SR_HARMONIZED`. Scenes with reported cloud percentage below 40% are retained, and the scene classification layer is used to mask cloud, shadow, defective, and no-data classes. Ten-day median composites are used to calculate NDVI, EVI, MSAVI, and NDWI:

$$NDVI = \frac{\rho_{B8} - \rho_{B4}}{\rho_{B8} + \rho_{B4}}, \tag{3}$$

$$EVI = 2.5\frac{\rho_{B8} - \rho_{B4}}{\rho_{B8} + 6\rho_{B4} - 7.5\rho_{B2} + 1}, \tag{4}$$

$$MSAVI = \frac{2\rho_{B8} + 1 - \sqrt{(2\rho_{B8} + 1)^2 - 8(\rho_{B8} - \rho_{B4})}}{2}, \tag{5}$$

$$NDWI = \frac{\rho_{B3} - \rho_{B8}}{\rho_{B3} + \rho_{B8}}. \tag{6}$$

Here, ρ_{B2}, ρ_{B3}, ρ_{B4}, and ρ_{B8} denote scaled blue, green, red, and near-infrared reflectance. NDWI is treated as a complementary water-related indicator rather than definitive evidence of inundation. Optical missingness caused by clouds or absent acquisitions is preserved.

3.6 CHIRPS precipitation and antecedent rainfall

Daily precipitation is obtained from `UCSB-CHG/CHIRPS/DAILY`. For each point and ten-day period, rainfall is aggregated to produce the ten-day total, maximum daily rainfall, number of rainy days, and antecedent accumulations such as 30-day and 90-day rainfall. For a window of k days:

$$P_{i,t}^{(k)} = \sum_{d=t-k+1}^{t} P_{i,d}, \qquad k \in \{30, 90\}. \tag{7}$$

A standardized precipitation index should only be reported if it is computed using a documented accumulation scale, reference period, and suitable statistical distribution. A simple standardized rainfall anomaly should not be described as a fully fitted SPI.

3.7 Candidate agricultural mask

Candidate agricultural pixels are derived from the `crops` class-probability band in Google Dynamic World (`GOOGLE/DYNAMICWORLD/V1`). For each year, the seasonal mean crop probability is calculated over 1 June–1 November. Pixels with mean crop probability at or above 0.30 are retained as a provisional agricultural mask:

$$C_{p,y} = \mathbb{I}\left(\overline{q}_{p,y}^{\text{crop}} \geq 0.30\right), \tag{8}$$

where $\overline{q}_{p,y}^{\text{crop}}$ is the seasonal mean probability and $\mathbb{I}$ is the indicator function. The threshold is a configurable starting value and must be assessed for sensitivity. Dynamic World probabilities indicate likely land-cover class, not crop species, crop calendar, or verified parcel boundaries. Agricultural exposure estimates are therefore conditional on the mask and its classification accuracy.

3.8 Area and temporal frequency of potential inundation

For a binary inundation mask $F_{p,t}$ and pixel area a_p in square metres, total potential inundated area for period t is calculated as

$$A_t^{\text{flood}} = \frac{1}{10^4} \sum_p a_p F_{p,t}, \tag{9}$$

where only valid radar pixels are included. The potential agricultural area exposed is

$$A_t^{\text{crop,flood}} = \frac{1}{10^4} \sum_p a_p F_{p,t} C_{p,y}. \tag{10}$$

The denominator for the exposed-cropland percentage includes only candidate crop pixels observed by valid radar in the corresponding period:

$$R_t^{\mathrm{crop}} = 100\,\frac{A_t^{\mathrm{crop,flood}}}{A_t^{\mathrm{crop,observed}}}. \tag{11}$$

A missing radar pixel is not counted as a non-flooded pixel. Seasonal frequency maps are computed from the number of valid periods with potential inundation divided by the number of valid radar observations at each pixel. Crop-exposure frequency uses the periods for which both the candidate crop mask and radar observation are valid. Connected-component counts, if reported, are treated as approximate and checked for sensitivity to spatial scale and component-size limits.

3.9 Next-period prediction target and leakage control

To prepare a forecasting dataset, the candidate target is the potential-inundation indicator for the next ten-day period:

$$Y_{i,t+1} = F_{i,t+1}. \tag{12}$$

The last period of each season has no next-period label and is excluded from supervised evaluation. Predictors must be restricted to information available at the forecast origin. Contemporaneous or future Sentinel-1 variables used to construct the target must not be included as unrestricted predictors in a model that claims to forecast inundation independently. Sentinel-2, rainfall, antecedent rainfall, and historical water context can be considered as predictors subject to timestamp checks and missing-data analysis.

The proposed chronological evaluation uses 2019–2023 for model development and 2024 as a held-out test season. Hyperparameter selection is performed only within the training years, preferably using year-blocked validation. If independent flood labels are unavailable, the task must be described as prediction of Sentinel-1-derived potential inundation, not prediction of independently verified floods.

3.10 Model candidates and evaluation

Five tabular classifiers are compared: a prior-prevalence dummy baseline; class-balanced logistic regression; class-balanced Random Forest (400 trees); XGBoost; and CatBoost. All use median imputation fitted on the training data. The 28 primary predictors include coordinates, calendar variables, radar and optical features, historical water extent, precipitation summaries, and radar-observation validity. The entirely missing `NDVI_ref`, identifiers, current flood label, next-period target, and `eau_ref_seche` are excluded. The latter is excluded because it contributes directly to construction of the flood label. A separate ablation measures the change when it is reintroduced. Hyperparameters are fixed in the script and not tuned on the 2024 test set. At a fixed classification threshold of 0.5, precision, recall, F1, balanced accuracy, ROC-AUC, average precision, and Brier score are reported.

Generalization is examined in three ways. First, five-fold grouped spatial validation is run on 2019–2023: the 200 sites are partitioned into five coordinate-based K-means

blocks, and each fold holds out all records from one block. The model is selected by mean fold average precision; per-block results and standard deviations are retained. Second, all models are trained on 2019–2023 and evaluated temporally on 2024, where sites recur from training. Third, a joint space–time evaluation predicts 2024 records only for a held-out spatial block, with that block excluded from all training years. This last result is the strictest available transfer test, although the five blocks are clusters rather than buffered spatial partitions. The target remains a Sentinel-1-derived proxy, not independently verified flood truth.

SHAP values are calculated for the model selected by spatial validation, on a random sample of up to 600 labeled 2024 observations with a training-derived background sample. Mean absolute SHAP values rank global contribution magnitude; summary and dependence plots display contribution direction and distribution. SHAP is an attribution of the fitted model, not a causal effect, and correlated predictors can share or redistribute importance [18].

Flood recurrence should be modeled as a count with the number of valid observations as exposure (e.g., Poisson or negative-binomial regression, with a hurdle model if excess zeros remain). Event-level area prediction has only 96 period records and is too small for a high-capacity learner. Cropland exposure requires numeric validated crop and flood masks; damage or yield-loss prediction additionally requires field damage classes, crop type/phenology, or independent yield references. Image segmentation (e.g., U-Net) is a future direction only after assembling independently labeled flood and cropland pixels. These are proposed extensions, not results of this experiment.

3.11 Validation and uncertainty

The VV threshold, dry-reference dates, Dynamic World crop-probability threshold, and spatial scale can influence estimated areas. Sensitivity analysis should vary these settings and compare mapped outputs with independent flood references where available. Spatially nearby points are not independent replicates; where feasible, validation should additionally group locations spatially. Cloud gaps, missing radar acquisitions, mixed pixels, seasonal water, irrigation, and land-cover classification errors are documented as sources of uncertainty.

4 Results

4.1 Dataset completeness and temporal coverage

The main export contains 19,200 rows for 200 sites, six years, and 16 ten-day periods per year. Each year–period group contains 200 records, site coordinates are constant within identifier, and no duplicate (`point_id`, `date_debut`) keys were found. There are 14,519 valid current flood labels (3.73% positive) and 13,583 valid next-period targets (3.76% positive); 29.26% of next-period labels are missing. Sentinel-1 fields and current labels are missing in 24.38% of rows, optical indices in 26.83%, and CHIRPS summaries in 0.50–1.16%. The `NDVI_ref` field is entirely missing. The next-period target was checked against a within-year, within-site one-period shift of the current flood label; all 13,583 comparable labels aligned.

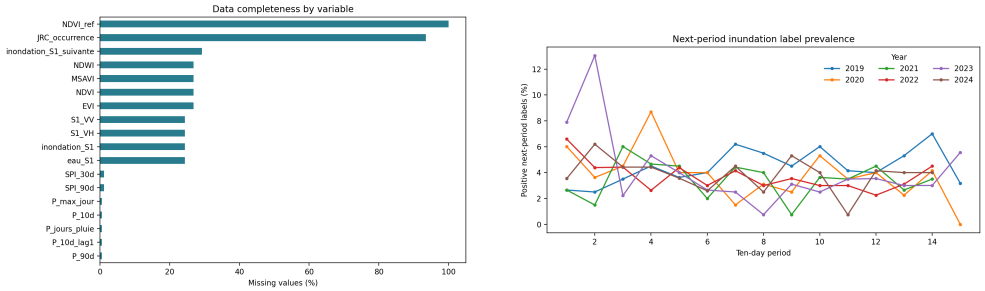


Fig. 2 Data missingness among the most incomplete variables (left) and observed prevalence of the next-period inundation target by ten-day period and year (right). The prevalence denominator includes only rows with a valid target.

Table 2 Period-level mapped inundated area and next-period target prevalence by year. Area values are not unique seasonal totals.

Year	Valid n	Flood target (%)	Mean/period (ha)	Max/period (ha)
2019	2,517	4.41	212,994	339,014
2020	2,337	3.72	148,678	300,576
2021	2,391	3.47	181,281	397,106
2022	2,111	3.60	142,964	220,249
2023	2,115	3.50	128,213	190,107
2024	2,112	3.79	128,055	276,264

4.2 Multiyear inundation dynamics

Across the 96 exported annual-period records, mean mapped area per period ranged from 128,055 ha in 2024 to 212,994 ha in 2019. The largest period-level estimate was 397,106 ha in 2021. The annual sum of these 16 period estimates is reported in the generated CSV as hectares-periods; it is not a unique seasonal footprint because the same land can be counted in multiple periods. Target prevalence varied from 3.47% in 2021 to 4.41% in 2019. Across all valid current labels, the mean sampled-point inundation frequency was 3.74%; the median site frequency was zero, while a few sites had much higher recurrence. Frequencies are conditional on valid observations and the fixed sample, not unbiased estimates for every location in the valley.

For the next-period target, the strongest pairwise rank association among available predictors was with JRC maximum water extent (Spearman $\rho = 0.287$); NDWI was positively associated ($\rho = 0.153$), while NDVI, EVI, and MSAVI associations were negative (absolute ρ approximately 0.160–0.163). The dry-reference water indicator was also associated with the target ($\rho = -0.234$), but is excluded from the primary model because it contributes to target construction. These coefficients are descriptive and not causal; inferential p-values are nominal with respect to repeated-site dependence.

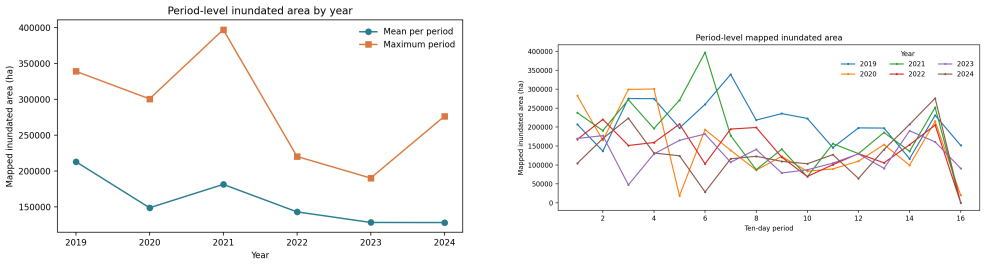


Fig. 3 Area summaries derived from the six period-level CSV exports: annual mean and maximum (left), and ten-day trajectories (right). These are mapped potential-inundation estimates, not field-validated flood extents.

Table 3 Five-fold spatial validation on 2019–2023. Values are mean (standard deviation) across the five held-out coordinate clusters.

Model	AP	AUC	Bal. acc.	Brier
Prevalence baseline	0.036 (0.019)	0.500 (0.000)	0.500 (0.000)	0.035 (0.018)
Balanced logistic regression	0.140 (0.144)	0.680 (0.119)	0.621 (0.089)	0.227 (0.071)
Balanced Random Forest	0.220 (0.214)	0.797 (0.065)	0.556 (0.080)	0.038 (0.013)
XGBoost	0.161 (0.136)	0.764 (0.072)	0.608 (0.099)	0.107 (0.093)
CatBoost	0.192 (0.178)	0.792 (0.077)	0.612 (0.105)	0.092 (0.078)

4.3 Model comparison and spatial validation

The 2019–2023 training data contain 11,471 valid next-period labels, including 431 positives. In five-fold site-grouped spatial validation with the label-overlapping dry-reference indicator excluded, the Random Forest was selected by mean fold average precision (0.220, standard deviation 0.214). CatBoost reached 0.192 (0.178), XGBoost 0.161 (0.136), logistic regression 0.140 (0.144), and the prevalence baseline 0.036 (0.019). Fold-to-fold variability is substantial, and one validation block contains only 33 positive labels; model ranking is therefore unstable.

4.4 Agricultural exposure

The supplied directory contains color-rendered GeoTIFF mosaics for inundation and agricultural exposure, but no numeric crop-exposure CSV or class-valued raster and no legend that permits a reproducible conversion of display colors to area. Consequently, agricultural area exposed and its percentage cannot be quantified from the available exports in this analysis. The point-level frequency map in Fig. 5 shows sampled flood recurrence only; it is not a cropland or damage map. No crop damage, yield reduction, or livelihood impact is measured here.

4.5 Prediction performance

Training used 11,471 labeled records from 2019–2023; the 2024 temporal test contains 2,112 records, including 80 positives. At threshold 0.5, the spatially selected Random

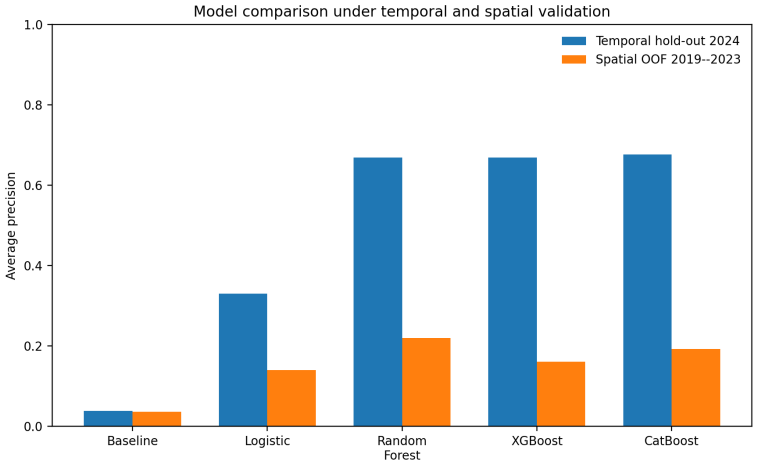


Fig. 4 Average precision by model under site-grouped spatial validation and the 2024 temporal hold-out. The 2024 scores are held out from model selection.

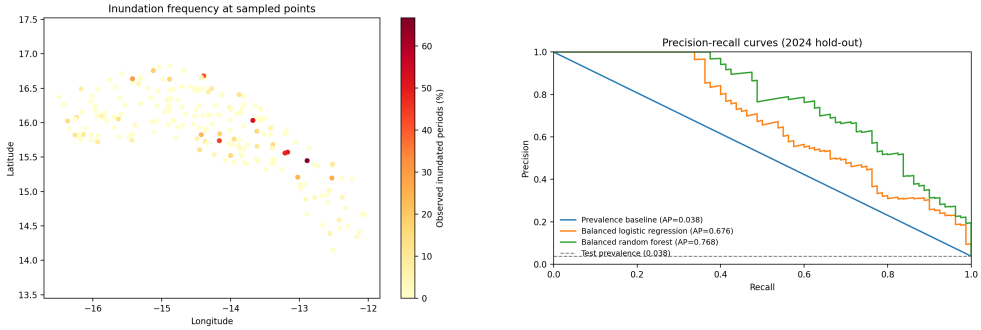


Fig. 5 Observed recurrence at the 200 sampled sites (left; valid periods only) and precision–recall curves for the held-out 2024 next-period classification task (right).

Forest has precision 0.868, recall 0.413, F1 0.559, balanced accuracy 0.705, ROC-AUC 0.943, average precision 0.669, and Brier score 0.023. CatBoost and XGBoost have higher average precision on the 2024 temporal test (0.677 and 0.669, respectively), but are not selected post hoc on that test set. The prevalence baseline has average precision 0.038, equal to test prevalence.

Joint space–time test. When 2024 sites are predicted only by models that excluded their entire spatial block from 2019–2023 training, the selected Random Forest obtains precision 0.750, recall 0.263, F1 0.389, balanced accuracy 0.630, ROC-AUC 0.835, average precision 0.379, and Brier score 0.037 on the 2,112 test records (80 positive). The lower average precision than the temporal-only score (0.669) quantifies the cost of transferring to both a future year and withheld locations.

Label-proxy ablation. For the selected Random Forest, reintroducing `eau_ref_seche`, which contributes to label construction, raises mean spatial-fold

Table 4 Exploratory model performance on the 2024 chronological hold-out set. The classification threshold is 0.5; average precision is the principal ranking metric under class imbalance.

Model	Prec.	Recall	F1	Bal. acc.	AUC	AP	Brier
Prevalence baseline	0.000	0.000	0.000	0.500	0.500	0.038	0.036
Balanced logistic regression	0.070	0.738	0.129	0.677	0.810	0.330	0.228
Balanced Random Forest	0.868	0.413	0.559	0.705	0.943	0.669	0.023
XGBoost	0.296	0.825	0.436	0.874	0.953	0.669	0.057
CatBoost	0.368	0.713	0.485	0.832	0.952	0.677	0.045

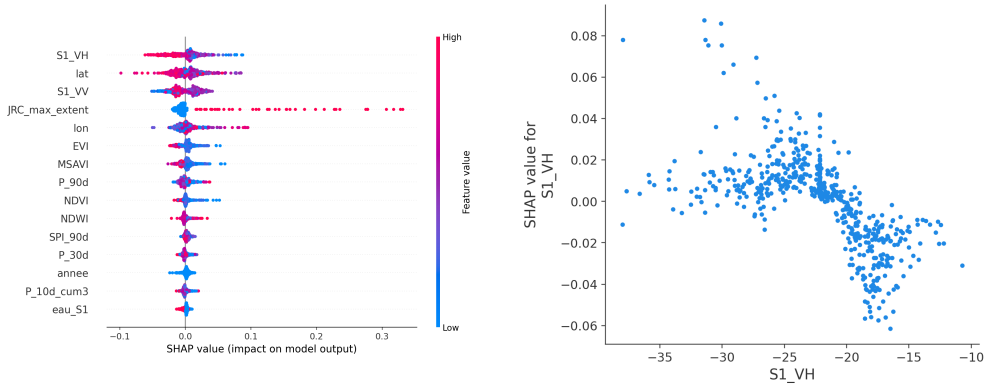


Fig. 6 SHAP explanation of the selected Random Forest without the label-overlapping dry-reference indicator. The left panel ranks global mean absolute contributions; the right panel shows the dependence of the most influential predictor, Sentinel-1 VH. SHAP attributions describe model behavior, not causal effects.

average precision from 0.220 (SD 0.214) to 0.425 (SD 0.212) and temporal 2024 average precision from 0.669 to 0.757. The primary metrics and SHAP analysis therefore exclude this variable; the ablation demonstrates the inflation associated with a predictor that is not independent of the target definition.

The largest mean absolute SHAP contributions are for S1 VH (0.0160), latitude (0.0160), S1 VV (0.0157), and JRC maximum water extent (0.0154), followed by longitude and optical indices. The close contributions and the prominence of coordinates and historical water indicate that geography and spatially structured prevalence remain influential. Pearson and Spearman tests similarly identify JRC maximum extent as the strongest available association with the next-period target ($r = \rho = 0.287$); correlations do not establish causal influence.

4.6 Comparison with published state of the art

The numerical comparison is contextual because label, spatial unit, lead time, and reference data differ. Nevo et al. report median inundation-extent F1 values of 0.69 under one-year cross-validation and 0.76 under leave-extreme-out testing for operational river-flood mapping; Nearing et al. report reliable forecasts of extreme river

Table 5 Contextual comparison with relevant published systems. Scores are not directly comparable across targets or reference data.

Study	Prediction target	Validation and reported result	Relation to this study
Nevo et al. (2022)	River stage and inundation extent at operational gauges	Year-held-out and leave-extreme-out; median inundation F1 0.69 and 0.76	Gauge/event maps, not point-level next-dekad labels.
Nearing et al. (2024)	Extreme river floods in ungauged watersheds	Up to 5-day lead time; reliability comparable to or better than GloFAS nowcasts	Global hydrologic warnings, not satellite-derived point classification.
This study, spatial hold-out	Next-period Sentinel-1 proxy at 200 sites	5-fold spatial AP 0.220 (SD 0.214); joint 2024 space–time AP 0.379	Valley-specific, sparse sample; label is not independently validated.

floods at up to five days lead time in ungauged watersheds relative to a global-system nowcast [9, 10]. Neither evaluates a ten-day, point-level Sentinel-1-derived binary label in the Senegal River Valley.

Accordingly, this work does not claim to outperform the global or operational state of the art. Its contribution is a documented multiyear point–period dataset, an explicit label-circularity test, spatial and temporal generalization results for this valley, and a reproducible analysis of candidate flood predictors. A direct performance claim would require common independent flood-event labels, matching lead times and spatial units, and a shared test protocol.

5 Discussion

The five-model comparison demonstrates why spatial hold-out matters: the Random Forest selected by historical spatial validation has spatial average precision 0.220 (SD 0.214), while its 2024 temporal score is 0.669 and its joint space–time score is 0.379. These results do not establish operational skill, and the high fold variability cautions against overinterpreting the selected model. The ablation shows that retaining `eau_ref_seche`, which contributes to label construction, materially inflates scores; it is excluded from the primary model. SHAP assigns contributions to radar, historical-water, optical, and geographic variables, but its geographic attributions may encode spatial prevalence and are not causal explanations. Published systems such as Nevo et al. and Nearing et al. predict different hydrological targets at different lead times, so their results contextualize the state of the art but cannot be used to claim a direct metric advantage. Period-level area sums double-count repeated inundation. The rendered exposure mosaics are not auditable numeric crop masks, and no field damage or yield outcomes are available.

6 Limitations

The workflow uses a fixed VV threshold and a pre-season water reference, which may misclassify smooth bare soil, permanent or seasonal water, and water under vegetation. Dynamic World probabilities are not field-verified crop parcels. Although the 200 identifiers and coordinates are stable, the point-generation code and sampling design were not supplied; these points cannot support unbiased valley-wide area estimates without a documented design and spatial weighting. Sensor missingness may not be random. The five coordinate clusters do not include spatial buffers, and one held-out block contains only 33 positive historical labels. The 2024 temporal test shares sites with training, while the joint space–time test has only 80 positives; uncertainty remains high. Pearson/Spearman p-values are based on repeated rows and may be anti-conservative despite multiple-testing correction. The target is a satellite-derived proxy, with a predictor-label relationship demonstrated by the dry-reference ablation; it is not an independent flood inventory. The RGB exposure mosaics and supplied CSVs do not yield a validated numeric agricultural-exposure area. Inundation overlap indicates exposure, not damage severity or yield loss; those outcomes require field surveys, crop-specific yield data, or validated crop-growth indicators.

7 Conclusion

This analysis provides an auditable EDA and five-model comparison for a 19,200-row Sentinel-derived point–period dataset in the Senegal River Valley. After excluding a label-overlapping predictor, the selected Random Forest achieves AP 0.220 in grouped spatial validation, 0.669 in a 2024 temporal hold-out, and 0.379 in a joint future-and-unseen-site test. These results are not directly comparable to operational river-stage or global extreme-flood systems and do not establish SOTA performance. Numeric cropland exposure, crop damage, and yield loss cannot be estimated from the available exports. Future work should add the original sampling/extraction scripts, independent flood labels, numerical validated crop masks, spatially blocked validation with buffers, and field/yield observations.

Declarations

Funding

The authors declare that no specific funding statement has been finalized for this manuscript.

Competing interests

The author declares no competing interests.

Data availability

The analysis used the supplied 2019–2024 CSV exports. Derived tables, model predictions, correlation outputs, SHAP values, and figures are saved in the accompanying

`resultats_inondations` directory. The source satellite products and Earth Engine assets remain subject to their respective access and sharing conditions; the point asset and extraction code should be archived with a persistent repository link before submission.

Code availability

The complete analysis and model-comparison script is `analyse_complete.py`; it calls `analyse_exploratoire_modeles.py` for shared EDA routines. The annual agricultural impact export is implemented in `export_impact_agricole_gee.js`; it creates six Google Drive CSV tasks. After downloading those six yearly CSVs to `article/data_gee_before`, run `fusionner_impact_agricole.py` to create the combined 2019–2024 table. The BibTeX references are in `references.bib`. The source point asset is external to this workspace, and its generation/sampling design code remains to be archived.

Author contributions

The author list includes Professor Cherif Bachir Deme, Head of the SATIC Department; Professor Adrien Basse of SATIC; and Professor Diery Ngom, Vice-Director of the UFR SATIC, Université Alioune Diop de Bambey. Individual CRediT contributions remain to be confirmed by all authors.

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